



The Global Equilibrium Real Interest Rate

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The Global Equilibrium Real Interest Rate: Concepts, Estimates, and Challenges

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Keywords

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Abstract

Real interest rates have been persistently below historical norms over the past decade, leading economists and policy makers to view the equilibrium real interest rate as likely to be low for some time. Various definitions and approaches to estimating the equilibrium real interest rate are examined, including approaches based on the term structure of interest rates and small macroeconomic models. The individual country approaches common in the literature are extended to allow for global trend and cyclical factors. The analysis finds that global factors dominate the downward trend in the equilibrium interest rate across 13 advanced economies. A corollary of this finding is that the U.S. equilibrium rate can be informed by global developments and is recently lower than estimated in U.S.-only studies. The analysis also highlights how the common global trend confounds empirical assessments of the determinants of movements in the equilibrium rate and the need to better integrate term-structure and macroeconomic approaches.

1. INTRODUCTION

Economists and financial market participants have gained conviction that the level of real (inflation-adjusted) interest rates likely to prevail over the long run—the equilibrium real interest rate—has declined to a level below the norms experienced over most of the past half century. These developments are global in scope, and research on the equilibrium real interest rate has had significant impacts on monetary policy discussions at the European Central Bank (e.g., Constâncio 2016) and at the Federal Reserve (e.g., Powell 2018).

The low level of the equilibrium real interest rate has been explored using tools from macroeconomics and finance. The connections—and the gaps—between the macroeconomic and financial approaches remain underdeveloped. This analysis considers the macroeconomic and financial approaches from a global perspective. Both approaches identify a substantial decline in the level of real interest rates likely to prevail over the long run owing to a common global factor. The global approach reveals a lower level of the equilibrium real interest rate in the United States than revealed by the typical individual country approach, reflecting the low level of interest rates among advanced economies. However, substantial differences in the level of the equilibrium rate are estimated by different methods, and uncertainty about the level is considerable. In addition, the drivers of the decline in the equilibrium rate are difficult to identify. The results point to a need for further integration of financial and macroeconomic approaches to understand trends in interest rates.

Following recent research, the equilibrium real interest rate is defined herein as a long-run notion, consistent with the equilibration of savings and investment over periods beyond the business cycle frequency. This definition more generally implies that the equilibrium real interest rate is the real interest rate consistent with macroeconomic balance over the long run, as highlighted in Section 2. The approach to estimation of the equilibrium real interest rate introduced by Laubach & Williams (2003) builds on the intuition of macroeconomic balance, extracting the trend in the real interest rate consistent with economic activity equal to its long-run potential level.

The analysis herein focuses on econometric approaches to estimating the equilibrium interest rate and its determinants, as outlined in Sections 3 and 4. One approach estimates the equilibrium real interest rate using information from the term structure of interest rates (as in, e.g., Bomfim 2001; Bauer & Rudebusch 2020; Christensen & Rudebusch 2019; Del Negro et al. 2019). The macroeconomic approach uses information from the relationship between output and interest rates to estimate the equilibrium real interest rate (as in, e.g., Laubach & Williams 2003; Clark & Kozicki 2005; Mésonnier & Renne 2007; Lubik & Matthes 2015; Laubach & Williams 2016; Holston, Laubach & Williams 2017; Brand, Bielecki & Penalver 2018; Fries et al. 2018; Johannsen & Mertens 2018; Bielecki, Beyer & Wieland 2019; Brand & Mazelis 2019; Jordà & Taylor 2019; Lewis & Vazquez-Grande 2019; Kiley 2020).

Most earlier work proceeds on an individual country basis—even multicountry studies, such as Holston, Laubach & Williams (2017) and Jordà & Taylor (2019), estimate the equilibrium rate at an individual country level and subsequently only consider comovement. I estimate the equilibrium real interest from a global perspective—that is, the models herein explicitly incorporate global developments in estimation of trends for individual countries, rather than estimating trends for individual countries and then extracting common trends. Del Negro et al. (2019) also explicitly models a global trend in an approach using term-structure data; these authors do not consider the approach on the basis of the comovement of output and interest rates.

The global approach yields important insights. Across 13 advanced economies, the common world component explains the overwhelming majority of the decline in the equilibrium real interest rate. Moreover, the dominance of global factors suggests that research on the determinants

of the equilibrium real interest rate should focus on such factors to a larger degree, as in Rachel & Smith (2017), Del Negro et al. (2019), Obstfeld (2019), and Rachel & Summers (2019), as highlighted in Section 5.

Before turning to the technical analysis, the practical relevance of the decline in the equilibrium real interest rate for economic policy is considerable. With regard to monetary policy, a low level of the equilibrium real interest rate implies that economic stabilization is impeded by the effective lower bound on nominal interest rates and may point toward adjustments in policy strategy to mitigate such adverse effects. For example, recent research points to a substantial increase in the frequency that an effective lower bound on interest rates may bind and impede economic performance (e.g., Kiley & Roberts 2017; Bernanke, Kiley & Roberts 2019; Kiley 2020). At the same time, uncertainty about the equilibrium real interest rate implies it may be difficult to find the optimal, or even a very good, approach that relies on an understanding of the equilibrium rate's level (e.g., Orphanides & Williams 2002; Giammarioli & Valla 2004; Weber, Lemke & Worms 2007; Bielecki, Beyer & Wieland 2019). Beyond short-term interest rate policy, shifts in the equilibrium rate affect the term structure and assessments of term premiums and hence may influence assessments of the efficacy of quantitative easing. For example, analyses of the effects of quantitative easing on term premiums may be sensitive to how models of term premiums account for changes in the equilibrium rate (e.g., has quantitative easing since 2008 persistently depressed bond yields?).

Fiscal policy discussions should also account for the equilibrium real interest rate. A low equilibrium real interest rate alters the sustainability of alternative government debt paths and hence raises questions regarding the appropriate stance of fiscal policy, especially given potential constraints on monetary policy from the effective lower bound (e.g., Furman 2016). The implications of a low equilibrium real interest rate for fiscal policy have attracted less research attention as of this writing. Rachel & Summers (2019) point to the potential for expansionary fiscal policy to raise the equilibrium real interest rate. Blanchard (2019) examines the implications of a low equilibrium real interest rate for debt sustainability.

The decline in the equilibrium real interest rate also has important implications for financial stability. Persistently low interest rates may create financial fragility by straining financial sector profitability and by incentivizing leverage (e.g., Borio & Disyatat 2014; Claessens, Coleman & Donnelly 2016). In addition, a core area in financial stability assessments involves assessments of valuation pressures and risk-taking (e.g., Adrian, Covitz & Liang 2015; Aikman et al. 2017; Adrian et al. 2019). The possibility that the decline in the equilibrium real interest rate reflects an increased premium on safety (e.g., Caballero, Farhi & Gourinchas 2017; Del Negro et al. 2017) points to a decline in expected returns on government bonds and a smaller, or no, decline in expected returns on risky assets. All of these effects make assessments of risk-taking and valuations difficult.

The next section discusses the evolution of real interest rates over the past half century and defines the equilibrium real interest rate. In light of this definition, Section 3 reviews approaches to estimating the equilibrium real interest rate and presents estimates. Section 4 reviews possible explanations for the decline and presents some new evidence. Section 5 highlights key open challenges.

2. DEVELOPMENTS AND DEFINITIONS

2.1. Real Interest Rates over the Past Half Century

To set the stage, **Figure 1** provides a snapshot of global interest rate developments, focusing on 13 advanced economies: Australia, Canada, Denmark, France, Germany, Italy, Japan, South Korea,

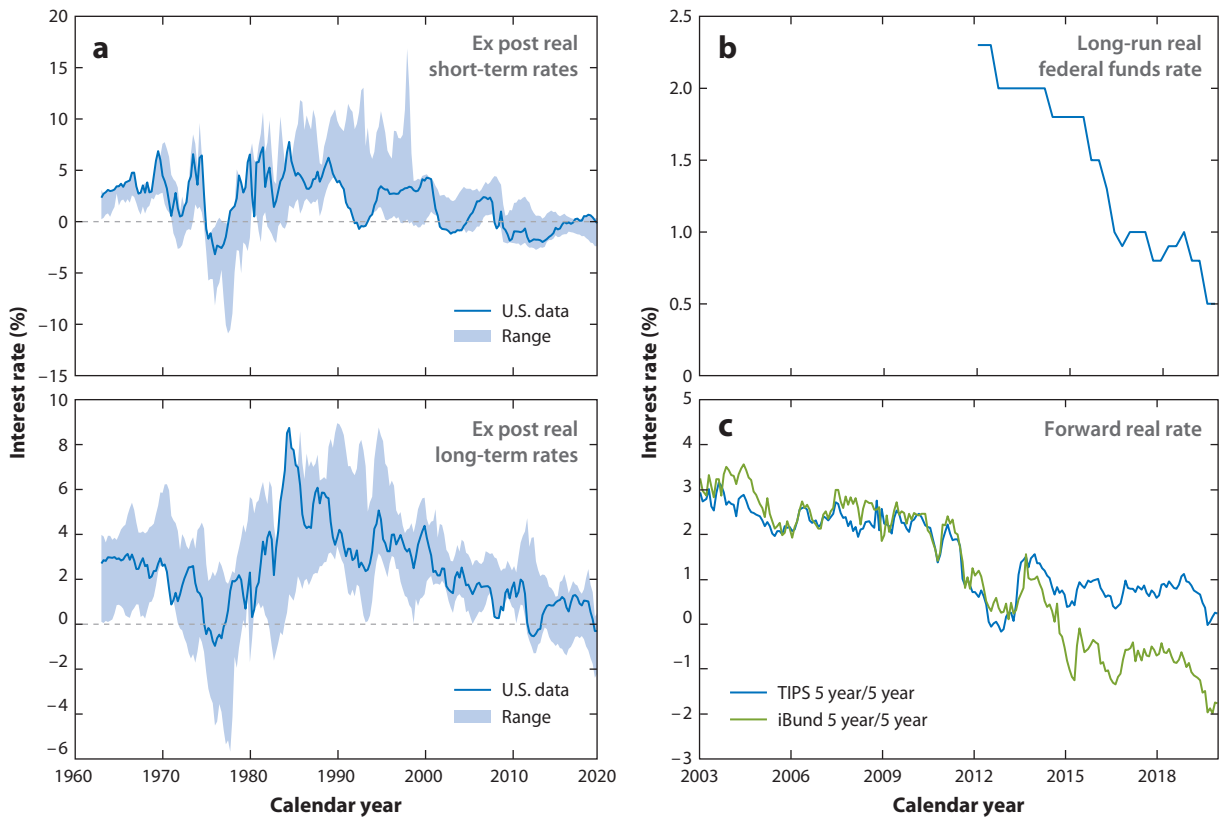


Figure 1

(a) Ex post real (top) short-term and (bottom) long-term interest rates across 13 economies. (b) U.S. Federal Open Market Committee participants' projection for long-run real federal funds rate. (c) Forward real interest rate implied by inflation-protected government bonds. Panel a data from the author's calculations and the Organisation for Economic Co-operation and Development (OECD 2019), obtained from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis. Panel b data from the author's calculations and the Federal Reserve Board, obtained from FRED at the Federal Reserve Bank of St. Louis. Panel c data from the author's calculations and the U.S. Treasury and Deutsche Bundesbank.

the Netherlands, Spain, Switzerland, the United States, and the United Kingdom. Real ex post short-term and long-term interest rates [constructed by subtracting consumer price inflation realized over the previous 3 years from a 3-month interbank (or related) interest rate or a 10-year government bond yield] have hovered in the neighborhood of 0% over the decade before 2019, substantially below the levels that prevailed over the previous 3 decades (Figure 1a).¹

Moreover, policy makers and market participants expect low real interest rates to persist. For example, participants in the Federal Open Market Committee, the monetary policymaking body of the U.S. Federal Reserve System, project that the long-run value of the real federal funds rate will equal approximately 0.5% (Figure 1b). In addition, forward real interest rates, as implied by inflation-protected government bonds in the United States and Germany, point to low, and even negative, real interest rates lasting beyond the next 5 years (Figure 1c).

¹The **Supplemental Appendix** provides details on data sources and the specific series used.

2.2. The Equilibrium Real Interest Rate (and the Natural Rate): Long Versus Short Run

To understand the low-frequency movements in real interest rates documented in **Figure 1**, researchers have focused on the equilibrium real interest rate. Three definitions are common. The equilibrium real interest rate is alternatively defined as

- the level to which real interest rates will converge in the long run absent shocks (e.g., Rachel & Smith 2017);
- the level of the real interest rate consistent with long-run equilibration of savings and investment (e.g., Obstfeld 2019); or
- the level of the real interest rate consistent with output equal to its long-run potential (e.g., Laubach & Williams 2003).

The equivalence of these definitions stems from common notions of macroeconomic balance. For example, short-run shifts in savings or investment demand (owing to, e.g., changes in consumer or business confidence) lead to transitory movements in real interest rates—movements that fade over time and are not associated with the drivers of lower-frequency movements in interest rates. In addition, these types of shifts drive business cycle movements in economic activity, pushing economic activity away from long-run potential. As a result, the definitions involving long-run equilibration of savings and investment and output equal to potential are closely linked. Finally, the first definition's focus on trend similarly abstracts from short-run considerations.

Despite their similarities, the alternative definitions are useful, as they have led researchers to develop different approaches to the estimation of the equilibrium real interest rate. The focus on the level to which real interest rates converge over the long run has led researchers to use techniques to estimate the equilibrium rate via simple trends or common long-run trends among interest rates (e.g., Del Negro et al. 2019). The definition involving the level of interest rates consistent with output at potential underlies the trend extraction model that dominates this literature, following Laubach & Williams (2003). And the approach focusing on equilibration of savings and investment drives analyses that use calibrated general equilibrium models to assess trends in the equilibrium real interest rate (e.g., Krueger & Ludwig 2007; Gagnon, Johansson & Lopez-Salido 2016; Rachel & Summers 2019).

The focus on the long run in discussions of the equilibrium rate distinguishes this concept from the Wicksellian notion of the natural rate of interest, which is typically defined as the level of the real interest rate that would equilibrate output to potential (and/or achieve price stability) in the short run (e.g., Woodford 2011). This concept fluctuates considerably in the short run, in response to business cycle shocks, and has been estimated in dynamic stochastic general equilibrium (DSGE) models (e.g., Edge, Kiley & Laforge 2008; Neiss & Nelson 2003). Kiley (2013) and Del Negro et al. (2017) discuss differences between equilibrium and natural rates in DSGE models.

3. ESTIMATION APPROACHES AND RESULTS

The approaches to estimating the equilibrium real interest rate vary somewhat, but each relies on the close connection between long-run equilibrium and reversion to trend over the long run. In essence, the various approaches span from simple-trend extraction to trend extraction informed by the economic mechanisms that inform researchers' understanding of real interest rate dynamics. Our discussion defers consideration of calibration approaches to Section 4.

3.1. Trend Extraction

The most direct approach to estimating the equilibrium real interest rate, or at least to informing assessments of a reasonable range for the level and variation in the equilibrium real interest rate, involves simple-trend measures. For example, Hamilton et al. (2016), Lunsford & West (2019), and Del Negro et al. (2019) consider moving averages over long time spans (e.g., 10 years). The fact that estimation of an underlying trend requires a moving average of such a long time period—combined with data availability typically spanning several decades to perhaps a bit more than 100 years—underscores how the data will have trouble informing assessments of the degree of variation in the equilibrium real interest rate (Lunsford & West 2019; Kiley 2020). For example, even the long macrohistory database of Jordà, Schularick & Taylor (2017) contains only 14 nonoverlapping decade observations.

The detrending approach followed herein involves an unobserved components model in which the short-term real interest rate fluctuates around an unobserved, random walk trend—similar to the approach to estimation of an inflation trend in Stock & Watson (2007) and to the approach to estimation of the equilibrium real interest rate used in Fiorentini et al. (2018). Specifically, the real short-term interest rate, $r^s(t)$, follows

$$\Delta r^s(t) = \rho \Delta r^s(t-1) + \lambda(r^*(t-1) - r^s(t-1)) + e(t), \quad 1.$$

$$r^*(t) = r^*(t-1) + v(t), \quad 2.$$

where $r^*(t)$ is the equilibrium real interest rate, $e(t)$ and $v(t)$ are independent and identically distributed (i.i.d.) shocks, and ρ and λ are parameters to be estimated. In this application, the parameters are estimated via Bayesian methods. The **Supplemental Appendix** provides estimation details. Important aspects of the Bayesian approach are relatively loose priors on all parameters except the standard deviation of shocks to the trend [$v(t)$]. Because real interest rates have varied relatively little over the past half century and previous work has found that estimation of the variance of the trend can be challenging (e.g., Del Negro et al. 2019; Kiley 2020), the prior distribution of the standard error of $v(t)$ is assumed to be an inverted gamma distribution with a mean of 0.25% (at an annual rate) and a standard deviation of 0.25%.²

3.2. Common Factor Among Interest Rates and Yield-Curve Models

The simple-trend approach abstracts from relationships among economic variables. Approaches that use the information from the yield curve exploit the connections implied by term-structure models. A parsimonious and flexible model of the yield on an N -period (real) bond, $r^N(t)$, is

$$r^N(t) = E(t) \left\{ \sum_{j=0}^{N-1} r^s(t+j) \right\} + \Xi(t), \quad 3.$$

$$\Xi(t) = A(L)X(t) + v(t), \quad 4.$$

where $E(t)$ is the expectations operator conditional on period t information. The first term in Equation 3 captures the expectations hypothesis of the term structure, and $\Xi(t)$ is a factor

²Kiley (2020) discusses the challenge associated with estimating the variance of the trend innovation and how this relates to the pileup problem associated with maximum likelihood estimation of the model of Laubach & Williams (2003).

governing deviations from the expectations hypothesis. Such deviations could be based on equilibrium models of the term structure that impose no-arbitrage restrictions (as in Kim & Wright 2005 and as discussed in Gürkaynak & Wright 2012) or on simple time series models. The central implications of this approach for estimation of the equilibrium real interest rate is that the short-term real rate $r^s(t)$ and the long-term real rate $r^N(t)$ (potentially for multiple N) share a common trend according to Equation 3, so long as $\Xi(t)$ is stationary.

In the related literature, authors follow different approaches. For example, Christensen & Rudebusch (2019) impose the absence of arbitrage and estimate a traditional term-structure model. In contrast, Del Negro et al. (2019) simply employ a common-trends model for short- and long-term real interest rates. Importantly, the simpler common-trends approach is an unrestricted version of the no-arbitrage approach (for cases in which the vector of factors is the same across models and as emphasized in, e.g., Cochrane 2007).

For this reason and in the interest of simplicity, our term structure-based approach models the 3-month real interest rate $r^s(t)$ and the 10-year real interest rate $r^N(t)$ using the common-trends specification. In contrast to the simple-trend approach, the model jointly considers developments across all 13 advanced economies. Denoting the 26×1 vector of real short- and long-term interest rates by $\begin{bmatrix} \Delta r^s(t) \\ \Delta r^N(t) \end{bmatrix}$, the model is given by

$$\begin{bmatrix} \Delta r^s(t) \\ \Delta r^N(t) \end{bmatrix} = P \begin{bmatrix} \Delta r^s(t-1) \\ \Delta r^N(t-1) \end{bmatrix} + \Lambda \left(SR^*(t-1) - \begin{bmatrix} r^s(t) \\ r^N(t) \end{bmatrix} \right) + E(t) + D \begin{bmatrix} c^s(t) \\ c^N(t) \end{bmatrix}. \quad 5.$$

In Equation 5, the equilibrium real interest rate $R^*(t)$, a 13×1 vector, is the common trend in $\begin{bmatrix} \Delta r^s(t) \\ \Delta r^N(t) \end{bmatrix}$ for each of the 13 countries, and the matrix S (of dimension 26×13) selects the appropriate country element of $R^*(t)$. $R^*(t)$ follows a random walk with country-specific shocks $V(t)$ (of dimension 13×1) and a global shock $w(t)$,

$$R^*(t) = R^*(t-1) + V(t) + w(t). \quad 6.$$

In addition and returning to Equation 5, $E(t)$ is a 26×1 vector of i.i.d. shocks, and $\begin{bmatrix} c^s(t) \\ c^N(t) \end{bmatrix}$ is a 2×1 vector of transitory global shocks to short- and long-term interest rates, with the matrix D (of dimension 26×1) selecting the appropriate element for each variable. $\begin{bmatrix} c^s(t) \\ c^N(t) \end{bmatrix}$ follows an autoregressive process of order 2. To reduce the number of parameters, the elements of P are restricted so that only own-country lags enter a country's equation, the elements of Λ are restricted so that error correction occurs between only the dependent variable and the appropriate country-specific trend, and the autoregressive processes for the common global cyclical components are univariate. As in the simple model above, estimation is via Bayesian methods with loose priors except on the standard errors of the innovations to $R^*(t)$. Relative to the simple-trend model, the term-structure model exploits information in both short- and long-term interest rates and includes global cyclical and trend factors.

A practical consideration taken herein, similar to the approach in much of the literature such as Del Negro et al. (2019), is the use of nominal yields and construction of real interest rates through subtraction of a proxy for inflation expectations (in which the proxy can be an estimate based on ad hoc smoothing of data or implied by a model). In recent decades, inflation-protected government securities have become common in advanced economies. Yields on such securities provide a direct measure of real interest rates and hence provide a natural set of data for analysis of the real yield curve (as discussed in, e.g., Gürkaynak & Wright 2012). However, data on real yields from inflation-protected securities are generally not available for long historical samples, and estimation of the underlying trend in interest rates requires such long samples. As a result, this approach is not explored herein—although some research, such as that by Christensen &

Rudebusch (2019) (building off Christensen, Lopez & Rudebusch 2010) has fruitfully pursued analyses with such data.

3.3. Small Macroeconomic Models

The term-structure model utilizes economic theory to a limited extent, as it is based on the notion that real short-term and long-term interest rates will share a common trend if term premiums are stationary. But this is a very limited reliance on theory relative to the definitions of the equilibrium real rate that motivate the analysis, which rely on notions of macroeconomic balance such as the long-run equilibration of savings and investment and the achievement of economic activity at potential levels.

The popular model of Laubach & Williams (2003) attempts to bridge the gap between motivation and application by exploiting the observed time series relationships between fluctuations of output around trend and interest rates.³ The model is based on the notion that cyclical movements in output are tied to cyclical movements in the real interest rate relative to the equilibrium interest rate—that is, the model is based on a standard IS curve from a textbook treatment (e.g., Woodford 2011).

The specification herein links the cyclical component of output, $[y(t)]$ (a 13×1 vector), to its own lags, a country-specific and global business cycle shock $[E(t)]$ and $c(t)$, respectively, and the short-term, real-interest-rate gap ($R^*(t - j - 1) - [r^s(t - j - 1)]$):

$$[y(t)] = \sum_{j=1}^2 \rho_j [y(t - j)] + \theta \sum_{j=1}^4 (R^*(t - j - 1) - [r^s(t - j - 1)]) + E(t) + Ic(t). \quad 7.$$

Supplemental Material >

θ (>0) is the slope of the IS curve—with higher interest rates $r^s(t)$, relative to the equilibrium level, depressing output. As in the term-structure model, the dynamics across countries are correlated through the cyclical global shock, $c(t)$, and there are no additional cross-country (short-run) dependencies. In addition, the equilibrium real interest rate has country-specific and global components and follows Equation 6. Finally, estimation is via Bayesian methods, with details provided in the **Supplemental Appendix**.

Focusing on the intuition, Equation 7 has two properties important for estimation of the equilibrium real interest rate. First, it has the standard trend-extraction property: If real interest rates are low, this likely reflects a mix of transitory shocks and declines in the equilibrium real interest rate. In addition, Equation 7 implies that the combination of low real interest rates and output below potential is particularly indicative of a low equilibrium real interest rate, as low interest rates are otherwise expected to boost output relative to potential.

3.4. Results

The analysis focuses on the 13 countries for which real interest rate data were presented in **Figure 1**: Australia, Canada, Denmark, France, Germany, Italy, Japan, South Korea, the Netherlands, Spain, Switzerland, the United States, and the United Kingdom. As is apparent from **Figure 1**, experience across these countries has varied considerably—with, for example, interest rates in many of these countries notably below U.S. levels in the years up to 2019. Details on sample period and estimation are provided in the **Supplemental Appendix**. In sum, the

³Laubach & Williams (2003) additionally include terms for inflation and, in some cases, the labor market; these additional terms aid identification of the business cycle component of output but are not critical to the intuition on how the model identifies the equilibrium real interest rate.

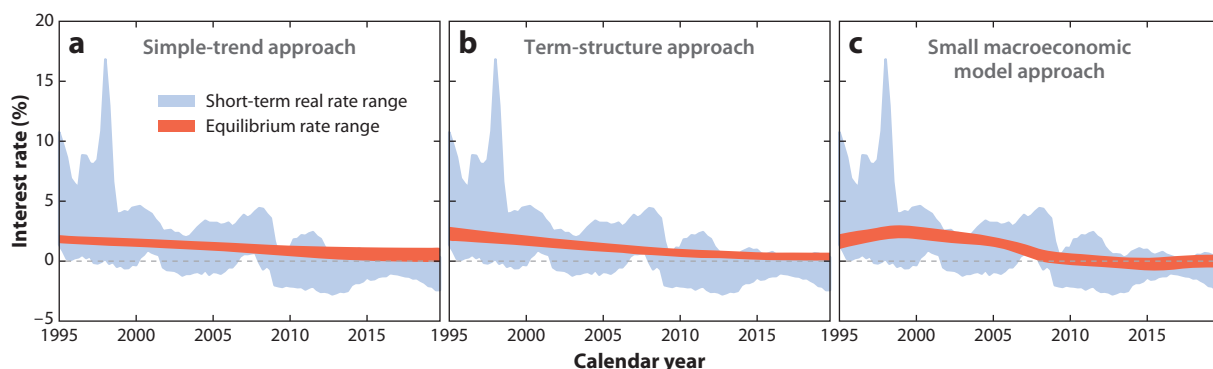


Figure 2

Ex post real short-term interest rates across 13 economies and the interquartile range of estimates of the equilibrium real interest rate. (a) Simple-trend approach. (b) Term-structure approach. (c) Small macroeconomic model approach. Data from the author's calculations and the Organisation for Economic Co-operation and Development (OECD 2019), obtained from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis.

sample roughly spans the mid-1960s to 2019, depending on country. For the small macroeconomic model approach, the cyclical component of output is obtained via the Hodrick & Prescott (1997) filter with a smoothing parameter of 140,000 (implying a smooth trend).

Figure 2 summarizes estimates of the equilibrium real interest rate for the last 25 years from each approach for the 13 countries. Two takeaways are apparent. All approaches show a decline in the equilibrium real interest rate. In addition, the decline is more significant for the two approaches that take a global perspective and use the entire set of countries to inform the estimate of the equilibrium real interest rate. For example, the simple approach—which is applied country-by-country—yields a decline in the central tendency of estimates (the median) from approximately 2% in the mid-1990s to a bit below 1% in 2019. The approaches of both the term-structure and simple macroeconomic models yield a decline in the median estimate of the equilibrium real interest rate from approximately 2% in the mid-1990s to approximately 0% in 2019.

Examination of the contribution of the global shocks to the evolution of the equilibrium real interest rate illustrates further the importance of a global perspective. **Figure 3** presents the short-term real interest rate for the United States along with the estimate of the equilibrium real interest rate and the contribution of the global shock to the equilibrium real interest rate for the past 40 years and for both the term-structure model and the small macroeconomic model. It is immediately apparent that the decline in the equilibrium real interest rate is primarily due to the global component.

Overall, the results on the importance of a global factor are intuitive and align closely with those of Del Negro et al. (2019). At the same time, the prominence of the global factor raises questions about the country-by-country approach that has dominated both the early and recent literatures.

Figure 3 also highlights another central result in the analysis herein and in the related literature: Estimates of the equilibrium real interest rate can differ notably depending on the model used. The term-structure approach estimates that the equilibrium real interest rate in the United States equals approximately 0.5% in 2019. The small macroeconomic model estimates that the equilibrium real interest rate equals approximately –0.5% in 2019.

Figure 4 further demonstrates the importance of global factors by comparing the estimate from the small macroeconomic model of the equilibrium real interest rate for the United States

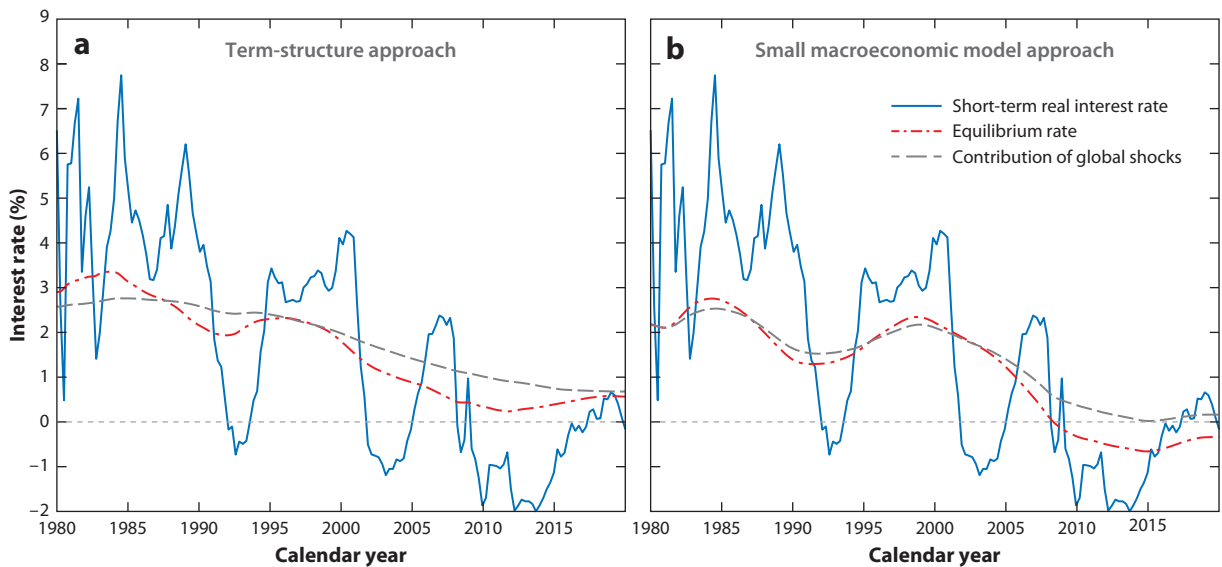


Figure 3

The ex post real short-term interest rate in the United States, the equilibrium real interest rate, and the contribution of the global shocks to the equilibrium real interest rate. (a) Term-structure approach. (b) Small macroeconomic model approach. Data from the author's calculations and the Organisation for Economic Co-operation and Development (OECD 2019), obtained from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis.

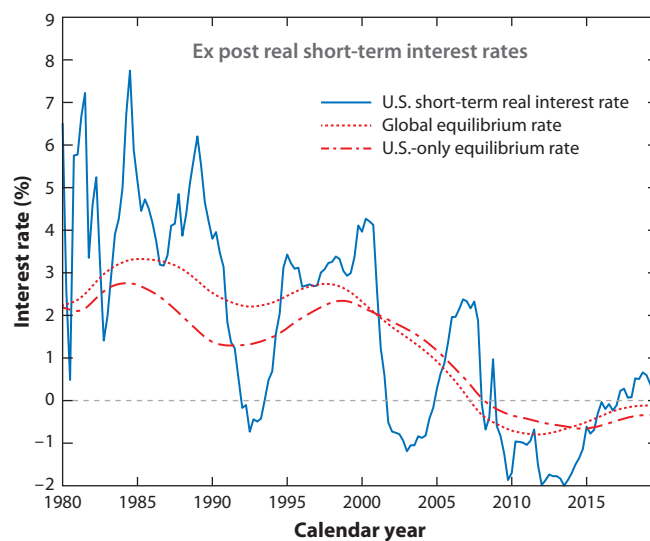


Figure 4

The ex post real short-term interest rate in the United States and the equilibrium real interest rate from global and U.S.-only models. Data from the author's calculations and the Organisation for Economic Co-operation and Development (OECD 2019), obtained from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis.

using global data with an estimate from the same specification (Equation 7) estimated using only U.S. data (and eliminating the global factors from the model and then re-estimating parameters). Although general trends are similar and point to a decline in the equilibrium rate, the variation in the equilibrium real interest rate is larger in the U.S.-only approach: The global model places the equilibrium real interest rate in the United States near 2% from 1980 to 2000 and subsequently lowers it to below 0%, whereas the U.S.-only approach places this concept close to 3% in the 1980s and fluctuates it more widely (as does the estimate from Laubach & Williams 2003). This finding is intuitive in light of global developments—combining information across countries provides greater information, and this information for other countries helps smooth the estimates of the equilibrium real interest rate in the United States.

More generally, a range of approaches suggests that the equilibrium real interest rate is likely notably below the norms of previous decades. These results are subject to considerable uncertainty, as discussed in Section 5. The next section reviews some of the possible sources of the decline in the equilibrium real interest rate, as these factors bear importantly on the policy questions.

4. DETERMINANTS OF THE EQUILIBRIUM REAL INTEREST RATE

4.1. Possible Determinants

An assessment of the factors contributing to decline in the equilibrium real interest rate requires a framework. Although the approaches in the literature vary, most analyses use a simple model of the supply and demand for loanable funds. The supply of loanable funds is an upward-sloping function of the interest rate, reflecting the decisions of households, businesses, and governments; loosely speaking, higher returns to saving boost the supply of savings. The demand for loanable funds is driven by the same set of actors and is downward sloping, reflecting the response of borrowers to higher costs of funds. **Figure 5** presents a simple figure summarizing equilibrium.

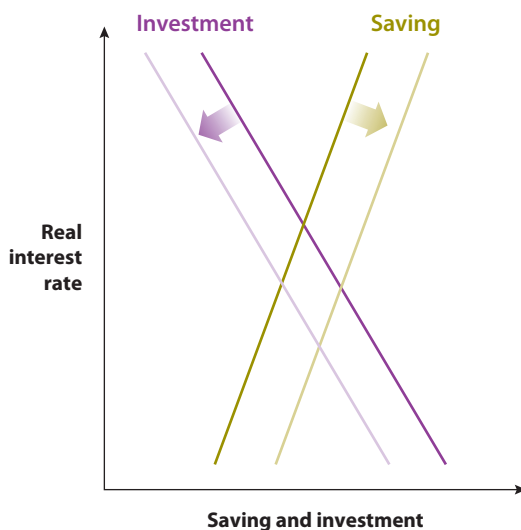


Figure 5

Saving and investment equilibrium. The figure illustrates hypothetical shifts in schedules for aggregate investment and savings. Investment demand is decreasing in the real interest rate, and savings demand is increasing in the real interest rate. A decrease in the investment demand schedule (*leftward shift*) or increase in the savings demand schedule (*rightward shift*) leads to a lower equilibrium real interest rate.

Before turning to evidence, a few important points from the literature can be gleaned from **Figure 5**. First, empirical evidence appears to suggest that both the savings and investment schedules are relatively inelastic (i.e., steep) with respect to interest rates. This property implies that observed shifts in the volume of savings and investment may be modest relative to the shift in interest rates. Rachel & Smith (2017) emphasize how the observed shifts in volumes of savings and investment over recent decades have been modest yet consistent with a potentially notable shift in the equilibrium real interest rate.

A second observation relates to the relevant market for loanable funds and the associated implications for empirical work. Many empirical exercises examine relationships between the determinants of savings and investment demand and trends in real interest rates at an individual country level (e.g., Lunsford & West 2019) or using a panel of countries (e.g., Hamilton et al. 2016; Ferrero, Gross & Neri 2017; Holston, Laubach & Williams 2017; Aksoy et al. 2019). But investment funds flow across borders and these flows have been viewed as key factors contributing to global trends in interest rates. For example, Bernanke (2005) introduced the idea that a global savings glut was an important factor contributing to low (long-term) interest rates in the mid-2000s. Caballero, Fahri & Gourinchas (2017), summarizing related research by these authors and others, point to increased savings in emerging markets and a related need for safe assets produced in advanced economies. These observations lead several analyses and discussions to focus on global factors (e.g., Rachel & Smith 2017; Obstfeld 2019; Rachel & Summers 2019). As our analysis below makes clear, it is empirically very challenging to discern the impact of country-specific and global factors, reflecting the high correlation between demographic and economic growth trends across advanced economies.

More generally, savings and investment depend on a large set of factors and hence the potential determinants of shifts in the equilibrium real interest rate are large. The sidebar titled *Factors Expected to Shift Aggregate Investment and Savings* summarizes key determinants emphasized in the literature, building on discussions in Rachel & Smith (2017), Lunsford & West (2019), and Rachel & Summers (2019)—each of which is based on overlapping-generation models of long-run economic growth. To be clear, each of the factors discussed is likely to affect both savings and investment, and the sidebar is meant to highlight whether forces are likely to shift primarily investment or savings. Starting first with factors influencing investment, slower growth, associated with either lower trend productivity growth or lower growth in the labor force, decreases required increases in the capital stock and consequently lower investment demand and interest rates.

Turning to savings, a range of social and policy factors are prominent in the literature. Focusing first on demographic factors, the dependency ratio (the fraction of the population below age 15 or above age 64) is likely inversely related to savings: A higher fraction of the population outside working age, and still consuming, should be associated with a lower saving rate. An increase in life expectancy, holding constant working years, is likely to raise aggregate saving as the working-age population boosts saving in anticipation of a longer retirement. Note that this effect is transitional and partial: An increase in life expectancy, holding constant years of work, eventually boosts the dependency ratio, which lowers savings. These effects occur over long periods and imply complex dynamics (e.g., Krueger & Ludwig 2007; Gagnon, Johannsen & Lopez-Salido 2016). Outside demographics, government saving is an important determinant of national and global saving (e.g., Elmendorf & Mankiw 1999). Finally, increased inequality is likely to boost savings and lower interest rates, as the propensity to consume of the rich is lower than that for other segments of the population (e.g., Dynan, Skinner & Zeldes 2004).

The bottom section of the sidebar titled *Factors Expected to Shift Aggregate Investment and Savings* focuses on factors that may affect the demand for saving in safe investment vehicles. This section contrasts with the top two-thirds of the sidebar, which focuses on overall savings and

FACTORS EXPECTED TO SHIFT AGGREGATE INVESTMENT AND SAVINGS

Factors Primarily Shifting Aggregate Investment

Economic growth

- Slower trend productivity growth decreases required increases in the capital stock and consequently lower investment demand and interest rates.
- Slower growth in the labor force decreases required increases in the capital stock and consequently lower investment demand and interest rates.

Factors Primarily Shifting Aggregate Savings

Demographic factors

- A higher fraction of the population outside working age—a higher dependency ratio—should be associated with a lower saving rate.
- An increase in life expectancy, holding constant working years, is likely to raise aggregate saving as the working-age population boosts saving in anticipation of a longer retirement.

Fiscal policy

- Government saving is an important determinant of national and global saving and hence influences interest rates.

Inequality

- Finally, increased inequality is likely to boost savings and lower interest rates, as the propensity to consume of the rich is lower than that for other segments of the population.

Factors Primarily Shifting the Demand for Safe Assets

- An increase in fundamental risk should depress the risk-free rate in a standard asset-pricing model.
- An increased demand for safe assets from emerging-market economies.
- Increased demand for safe assets driven by regulatory or other factors.

investment—and hence discusses the effects on interest rates, the marginal product of capital, and returns on risky assets. For example, an increase in fundamental risk should depress the risk-free rate in a standard asset-pricing model, without necessarily having similar effects on the marginal product of capital (e.g., Kiley 2004). The demand-for-safe-assets mechanisms in Caballero, Farhi & Gourinchas (2017) should disproportionately affect safe interest rates, resulting in wider spreads of required returns on risky assets over safe assets. Other models have similar implications based on alternative drivers (e.g., Del Negro et al. 2017). These mechanisms are important to keep in mind, as research has emphasized that the decline in real interest rates over recent decades has not been accompanied by declines in measures of the return to productive capital (e.g., Farhi & Gourio 2018), casting doubt on mechanisms in which movements in real interest rates and the return to capital are tightly linked.

4.2. Reduced-Form Regressions

Our empirical analysis of determinants follows recent analyses that relate measures of trends in real interest rates to some of the factors emphasized in the literature. The analysis considers only

Table 1 Estimates of the relationship between the 5-year moving average of real interest rates and determinants across 13 advanced economies

	Case 1	Case 2	Case 3	Case 4	Case 5
Dependency ratio	−0.07 (−0.22, 0.5)	NA	NA	−0.17 (−0.30, −0.07)	0.04 (−0.03, 0.11)
Life expectancy	NA	−0.27 (−0.55, 0.02)	NA	−0.32 (−0.54, −0.05)	0.49 (−0.11, 1.19)
GDP growth	NA	NA	0.45 (−0.09, 0.99)	0.38 (−0.17, 0.96)	0.28 (−0.19, 0.85)
Aggregate (population-weighted) dependency ratio	NA	NA	NA	NA	−0.31 (−1.87, 0.85)
Aggregate (population-weighted) life expectancy	NA	NA	NA	NA	−0.61 (−1.10, −0.27)
Aggregate (population-weighted) GDP growth	NA	NA	NA	NA	−1.09 (−2.12, −0.33)

Regressions include country-fixed effects. In each case, the independent variable in the regression is the 5-year moving average of the real short-term interest rate within a country. The independent variables entering the regression in each case are those for which coefficients are reported.

Abbreviations: GDP, gross domestic product; NA, not applicable.

a few potential determinants, for two simple reasons. First, the underlying trends in real interest rates largely consist of a downward movement since the mid-1980s (**Figure 1**). Many of the determinants similarly contain similar trends—with, for example, economic growth generally decelerating (on a trend basis) since that time and life expectancies generally increasing since the 1960s. These developments result in a high degree of correlation across determinants, making reduced-form empirical assessments challenging. In addition, data for an extended period on some potential determinants are not available.⁴

Following previous cross-country analyses, the empirical approach focuses on demographic factors and underlying growth trends. For demographic factors, the analysis considers the dependency ratio and average life expectancy at birth, both from the United Nations. For underlying growth trends, the 5-year moving average of gross domestic product (GDP) growth is used. Whereas other analyses have taken more complicated approaches, the equation herein relates the 5-year moving average of the short-term real interest rate within a country to the determinants via a simple fixed-effects panel regression. In addition to the country-specific variables, a specification including the population-weighted averages of the independent variables, in addition to their country-specific values, is also considered. **Table 1** reports results. Several results stand out. None of the potential determinants is significantly different from zero when considered in isolation (i.e., each 90% confidence interval contains 0). Joint consideration of the dependency ratio, life expectancy, and underlying growth finds statistically significant relationships among the underlying real short-term interest rate, the dependency ratio, and life expectancy. Life expectancy has the expected negative sign (as a longer life expectancy is expected to boost saving for retirement and thereby lower interest rates). However, the dependency ratio has the wrong sign. Note that Ferrero, Gross & Neri (2017) and Aksoy et al. (2019) find a similar correlation, and Aksoy et al. (2019) present a more complicated model in which a higher dependency ratio

⁴For example, the World Inequality Database (<https://wid.world/>) covers the countries we analyze, but basic inequality measures are missing for many countries before recent decades.

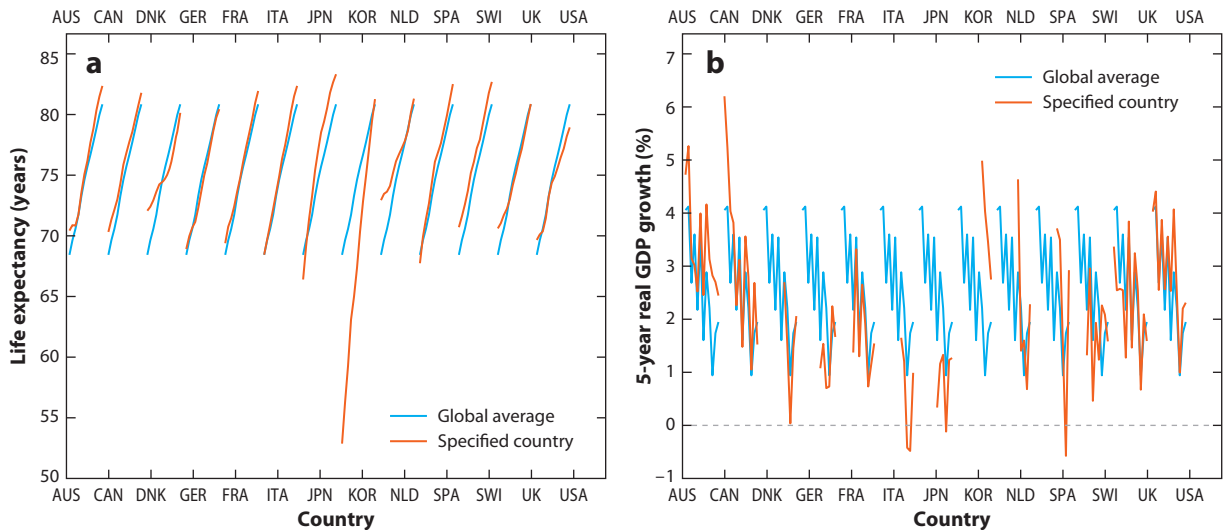


Figure 6

Trends in life expectancy and economic growth (1960–2015). (a) Life expectancy by global average and by specific country. (b) Five-year real gross domestic product (GDP) growth by global average and by specific country. Country abbreviations: AUS, Australia; CAN, Canada; DNK, Denmark; FRA, France; GER, Germany; ITA, Italy; JPN, Japan; KOR, South Korea; NLD, the Netherlands; SPA, Spain; SWI, Switzerland; UK, the United Kingdom; USA, the United States. Data from the author's calculations and the Organisation for Economic Co-operation and Development (OECD 2019), obtained from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis.

depresses trend innovation and growth and thereby depresses the real interest rate. Finally, adding the global averages results in most variables being insignificant.

This result is not surprising but highlights an important point. As emphasized in **Figure 3**, most of the decline in the equilibrium real interest rate appears to come from a common global factor that has fallen slowly—like a trend—over time. Many potential determinants of the equilibrium real interest rate, such as life expectancy, underlying growth, and inequality, similarly have exhibited high degrees of comovement dominated by a low frequency component over the past half century. As a result, empirical assessments are plagued by multicollinearity. This issue is illustrated in **Figure 6**, which plots the global average for life expectancy along with country experiences (**Figure 6a**) and the global average for 5-year GDP growth along with country experiences (**Figure 6b**). The similarity in patterns in the variables across time and countries is striking.

In addition to the econometric challenges associated with collinearity, the regression or correlation approach, while common, has conceptual issues. Most importantly, the steady-state or long-run relationships highlighted in the sidebar titled *Factors Expected to Shift Aggregate Investment and Savings* can result in complicated dynamics that play out over decades, reflecting life-cycle effects associated with demographic transitions and the slow movement in the capital stock in response to persistent shifts in savings and investment. As a result, research has considered implications from calibrated general equilibrium models.

4.3. Calibrated General Equilibrium Models

The most common factors considered in calibrated general equilibrium models are demographic factors such as shifts in the dependency ratio and life expectancy, modeled in overlapping-generation models of various degrees of complexity (e.g., Krueger & Ludwig 2007; Carvalho,

Ferrero & Nechio 2016; Gagnon, Johannsen & Lopez-Salido 2016; Bielecki, Brzoza-Brzezina & Kolasa 2018). In general, these models find substantial effects of demographic changes. For example, Krueger & Ludwig (2007), in an analysis that predates much of the recent literature documenting a decline in the equilibrium real interest rate, projected a decline in the equilibrium rate from the early 2000 levels of just below 1 percentage point. More recently, Carvalho, Ferrero & Nechio (2016) and Gagnon, Johannsen & Lopez-Salido (2016) both suggested that the aging of the baby boom generation and increases in life expectancy contribute to a decline in the equilibrium real interest rate from the mid-1980s to the present rate of approximately 1.5 percentage points. This estimate is a substantial share of the overall decline estimated herein and in related studies.

A more comprehensive assessment is reported by Rachel & Summers (2019)—although their estimates reflect an assembly of results from different models. Overall, they find a 1.7 percentage point decline in the equilibrium real interest rate. Demographic factors account for a 1.8 percentage point decline, in the range of Carvalho, Ferrero & Nechio (2016) and Gagnon, Johannsen & Lopez-Salido (2016). Importantly, Rachel & Summers (2019) also consider the effects of a decline in trend productivity growth and increased inequality. The combination of slower trend productivity growth and increased inequality is estimated to lower the equilibrium real interest rate by another 2.5 percentage points. Increased government debt and higher government spending on the elderly, the combination of which is estimated to raise the equilibrium real interest rate by 4 percentage points, offset these drags on the equilibrium real interest rate. Overall, the key takeaway from Rachel & Summers (2019) is that a model with relatively inelastic savings and investment schedules can imply large, and potentially offsetting, movements in real interest rates in response to changes in demographics, productivity, and government policies. Whether such large offsetting effects are a good description of the data merits further analysis.

In addition, it is noteworthy that the calibrated general equilibrium framework has focused on factors influencing savings and investment and hence capital accumulation. These mechanisms predict similar movements in real interest rates and the marginal (and under some conditions, average) product of capital. Measures of the average product of capital do not show signs of a strong downward trend (e.g., Farhi & Gourio 2018). The divergence between safe interest rates and broader measures of the return to capital points to a widening of risk spreads associated with demand for safe assets, not shifts in savings and investment, as a potential driver of the equilibrium real interest rate (e.g., Del Negro et al. 2017).⁵

5. OPEN QUESTIONS AND CHALLENGES

5.1. The Term-Structure and Macroeconomic Models

A striking feature of the estimates in Section 3, and in the literature, is the separation between the approach using information from the term-structure and the macroeconomic approach. This separation is clearly inconsistent with the notion that financial and macroeconomic conditions are jointly determined. But it is consistent with the common approaches taken in both the term-structure and macroeconomic literatures.

The role of macroeconomic factors in shaping the term structure of interest rates is a component of ongoing research in term-structure modeling (e.g., Gürkaynak & Wright 2012). Large

⁵Eggertsson, Mehrotra & Robbins (2019) suggest that increases in monopoly rents have accompanied the shifts in savings and explain the divergence between measures of the average product of capital and real interest rates.

macroeconomic models typically incorporate important effects of short- and long-term interest rates on economic activity (e.g., Boivin, Kiley & Mishkin 2010). Nonetheless, the types of models used to estimate the equilibrium real interest rate from a macroeconomic or term-structure perspective have largely proceeded independently.⁶

Progress toward ameliorating this situation depends on developments in (at least) two areas. First, the macroeconomic literature has made limited progress in developing parsimonious models in which long- and short-term interest rates affect economic activity (e.g., Kiley 2014). Such progress is a precondition for joint determination of economic activity, short-term interest rates, and long-term interest rates—and hence a precondition for models that combine the macroeconomic and term-structure approaches.

Second, our understanding of the term structure needs to account for the nonstationarities induced in nominal interest rates by trends in inflation (e.g., Stock & Watson 2007) and the equilibrium real interest rate. Traditional term-structure models assume nominal, or at least real, interest rates are stationary (e.g., Kim & Wright 2005), with only limited exceptions (e.g., Christensen & Rudebusch 2019).⁷ In addition to facilitating integration of term-structure approaches to estimation of the equilibrium real interest rate, accounting for trends in inflation and the equilibrium real interest rate may shift understanding of term-structure dynamics.

For example, Davis, Fuenzalida & Taylor (2019) consider an approach to integrate the type of variation in the equilibrium real interest rate modeled in the small macroeconomic model approach of Laubach & Williams (2003) with standard term-structure models. Their analysis highlights the importance of including trends in the equilibrium real interest rate and inflation in term-structure models for understanding drivers of yields, and it further illustrates the importance of such considerations for an assessment of bond risk premia. At the same time, their analysis does not consider the links between the business cycle and interest rates directly and hence is just one example of the direction of work needed to unify the macroeconomic and term-structure approaches.

5.2. Uncertainty Regarding the Level and Determinants of the Equilibrium Rate

A high degree of uncertainty surrounds the equilibrium rate. For example, our discussion of the determinants of the equilibrium real interest rate highlighted how short data samples, substantial collinearity, and the potential for a multitude of offsetting effects cloud understanding. Moreover, our presentation of estimates of the equilibrium real interest rate highlighted how different approaches can provide quite different estimates. Overall, uncertainty is high, and this uncertainty has led many to question how discussions of the equilibrium rate enter policy discussions (e.g., Orphanides & Williams 2002; Weber, Lemke & Worms 2007; Bielecki, Beyer & Wieland 2019).

The fact that the equilibrium real interest rate is unobserved and evolves slowly implies that uncertainty regarding its value—especially its contemporaneous value—is an irreducible feature. This uncertainty has important practical consequences, illustrated in **Figure 7**. First, estimates of the equilibrium rate based on data up to a given period are revised substantially as more data arrive. This is shown in **Figure 7a**, which plots the one-sided estimate of the equilibrium real

⁶Johannsen & Mertens (2018) use term-structure information in a small macroeconomic model of the equilibrium real interest rate for the United States but do not consider the effects of both long- and short-term interest rates on economic activity.

⁷Given that the equilibrium real interest rate focuses on the long run, approaches that use survey estimates of long-run expected inflation, interest rates, and economic growth could be informative. These approaches are not common in the literature on the equilibrium real interest rate. Kim & Wright (2005) and related term-structure models have used survey expectations.

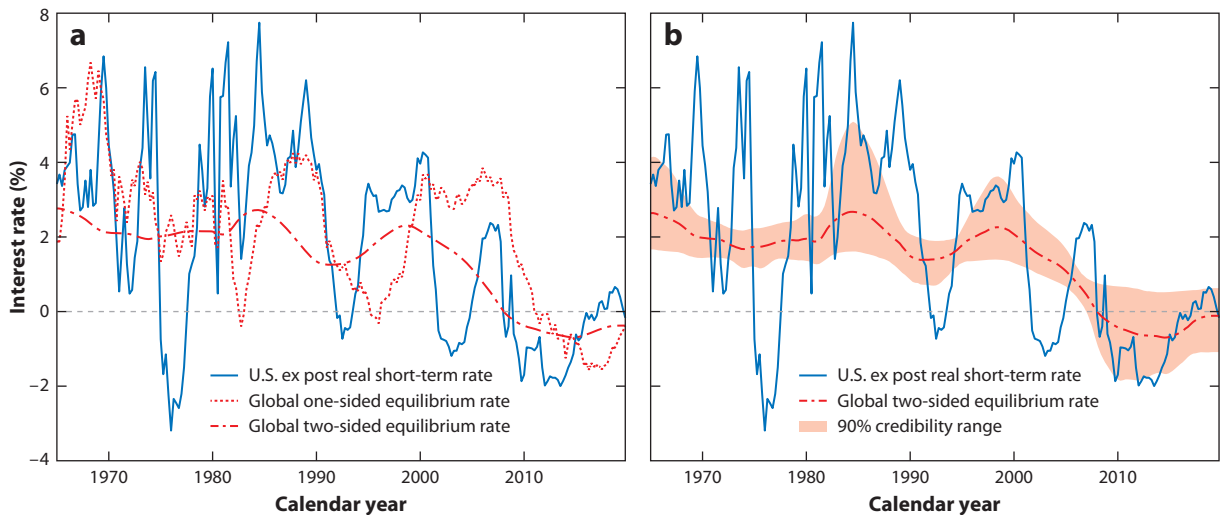


Figure 7

Uncertainty about the equilibrium real interest rate. (a) The ex post real short-term interest rate in the United States and the one-sided and two-sided estimates of equilibrium real interest rate from the global macroeconomic model. (b) The two-sided estimates of equilibrium real interest rate from the global macroeconomic model and the 90% credible interval. Data from the author's calculations and the Organisation for Economic Co-operation and Development (OECD 2019), obtained from Federal Reserve Economic Data (FRED) at the Federal Reserve Bank of St. Louis.

interest rate from the global small macroeconomic model along with the estimate that uses all available data. (Note that the last observation of these two concepts is identical by definition.) It is clear that a researcher or policy maker operating in real time would have perceived the equilibrium real interest rate to be substantially higher than it is currently estimated to have been in, for example, the 2000s; this impression is confirmed by examination of Federal Reserve staff estimates of the equilibrium real interest rate reported in staff forecast materials for the mid-2000s. Indeed, **Figure 7** makes clear that the evidence for a decline in the equilibrium real interest rate within this model essentially appeared in the wake of the global financial crisis of 2008, as the one-sided estimate places the equilibrium real interest rate near 3% in 2008. The magnitude of these revisions drives home the challenges for policy analysis raised in Weber, Lemke & Worms (2007) and Bielecki, Beyer & Wieland (2019).

Second, the credible interval surrounding estimates of the equilibrium real interest rate is wide, especially at the end of a data sample (as shown in **Figure 7b**). Even so, the top of the 90% credible interval is only 3/4% for the United States in 2019.

6. SUMMARY

Policy makers and economists have taken on board the persistence of low real interest rates and widely view the equilibrium real interest rate as low by historical standards. A range of approaches to estimating the equilibrium real interest rate confirm a pronounced downward trend among advanced economies in the level of real short-term interest rates likely to prevail over the longer term. Several areas deserve further scrutiny.

First, the analysis herein extended the dominant country-by-country approach to estimation of the equilibrium real interest rate to a global perspective. A joint global model of the term-structure or of the macroeconomic approach shows that global factors overwhelmingly drive the

decline in the equilibrium real interest rate. This emphasizes that the source of the decline is likely to be global, suggesting that the econometric analyses of the determinants of the equilibrium real interest rate that use panels of countries may need to consider alternative approaches. The global approach also suggests that the equilibrium real interest rate in the United States may be lower than appreciated, reflecting the low level of interest rates among advanced economies relative to U.S. levels.

In addition, the term-structure and macroeconomic approaches to estimation of the equilibrium real interest rate have largely proceeded along parallel tracks, and it would be useful to see feedback among these workstreams. This feedback requires models of the term structure to incorporate trends in interest rates. It also requires richer macroeconomic models of how the term structure of interest rates affects economic activity.

Finally, the equilibrium real interest rate is very difficult to estimate in real time. Although all indications point to a level of the equilibrium real interest rate that was very low by historical standards in the late 2010s, researchers and policy makers should be prepared to revise their views—in whatever direction the data take them—in coming years. A look at estimates of the equilibrium real interest rate using data through 2008, which point to much higher values for the equilibrium real interest rate than are estimated for that year using data through 2019, illustrates the potential for large changes in assessments.

DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review. The views expressed in this work are those of the author and do not represent the views of the Federal Reserve Board or its staff.

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